Lecture 14
Time Series Data and Markov Models

Algorithms, AI & Data Science II

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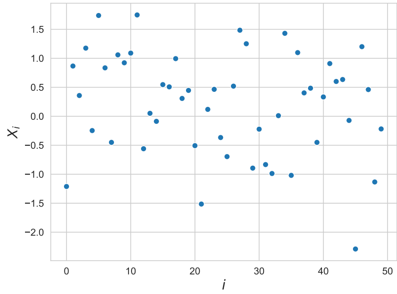
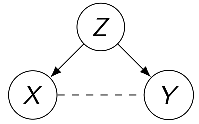
Motivation

- we introduced techniques for **statistical modelling and inference** in empirical data
- we considered causal diagrams, where nodes are **random variable** and edges represent **causal relationships or correlations**
- example for **probabilistic graphical model** that can be used to represent **relationships in data**
- so far, we considered sets of **independent realizations X_i** of random variable
- how can we model **time series data**, i.e. values X_t recorded at different times t



50 independent realization X_i of random variable $X \sim$ Norm



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- **Lecture L14:** Time Series Data and Markov Models 25.06.2025
- **Educational Objective:** We discuss statistical models for time series data and introduce Markov chains to model categorical sequences. Exploring random walks in graphs, we clarify under which conditions Markov chains reach a stationary state.
  - Stochastic Processes and Markov chains
  - Random Walks in Graphs
  - Stationary States of Markov Chains
- **Handout version:** *2025/06/23 10:18:06*

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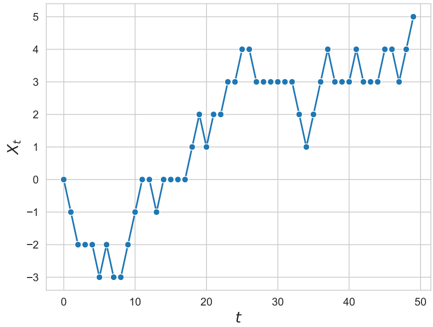
Stochastic processes

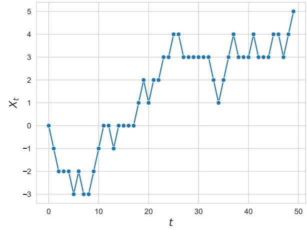
- consider **time series data** X_t with time index t
- we call collection of random variables $X_t \in E$ a **stochastic process** with state space E
- we can have **discrete or continuous state space**, e.g. $X_t \in \mathbb{R}$ or $X_t \in \{a, b, c\}$
- we can have **discrete or continuous time**, e.g. $t \in [0, \infty)$ or $t \in \{0, 1, 2, \ldots\}$
- how can we **model patterns** in such stochastic processes?
- we can **use probabilities to model uncertainty about future states** X_t of the process



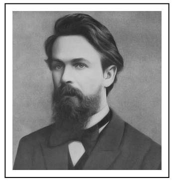
discrete-time stochastic process with **discrete state space**

Markov models

- we can use **probabilities** to model **stochastic processes** and resulting **time series data**
- assign $P(X_t)$ of **future states**, possibly conditional on current/past states
- Markov models assume that system **memoryless**, i.e. next state only depends on current state and not on past states
  - continuous-time $\Rightarrow$ **Markov process**
  - discrete-time $\Rightarrow$ **Markov chain**
- important foundation for **stochastic modelling of time series data**
- numerous applications in **pattern recognition, time series forecasting, recommender systems**



discrete-time process with **discrete state space**



Andrey Markov

1856 – 1922

image credit: Wikimedia Commons, public domain

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Markov chains

- consider **discrete-time** stochastic process on **finite, discrete state space** where X_t is state of process at time $t \in \{0, 1, \dots\}$
- $P(X_t = s)$ models uncertainty about state $s \in E$ of process at time t
- at time t **transition probability** from state s_t to state s_{t+1} can be given as

$$P(X_{t+1} = s_{t+1} | X_t = s_t, X_{t-1} = s_{t-1}, \dots X_0 = s_0)$$

- process is **memoryless**, i.e. has the **Markov property**, iff

$$P(X_{t+1} = s_{t+1} | X_t = s_t) = P(X_{t+1} = s_{t+1} | X_t = s_t, X_{t-1} = s_{t-1}, \dots X_0 = s_0)$$

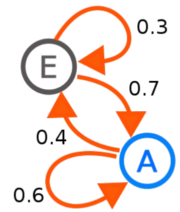
  for all sequences of states $s_0, \dots, s_{t-1} \in E$
- we obtain a **Markov chain model** with discrete and finite state space

observations

- next state X_{t+1} only depends on current state X_t
- transition probabilities between states s_i and s_j can be given in entries T_{ij} of **transition matrix** $\mathbf{T} \in \mathbb{R}^{n \times n}$ where $n = |E|$

Applications of Markov chains

- Markov chains are important class of **probabilistic graphical models**
- nodes of graph represent **states of a system**, while edges represent **transitions between states**
- assuming model for transitions, Markov chains allow to answer questions about **dynamic behavior of systems**



graph representation of Markov chain with two states A and E

exemplary questions

- what is the **probability that system is in given state** at given point in time?
- how long will it take on average **to first reach a given state**?
- does system reach a **stationary state** in which the probabilities of states seize to change?

exemplary applications of Markov chains

- packet arrivals in communication networks
- user click streams in information systems
- diffusion processes in networks
- search and rescue operations

- **important applications across disciplines**

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Example application

exemplary "search and rescue" application

- you loose your drunk friend in the streets of Würzburg
- you need to search for him/her?

- **model Würzburg as graph** $G = (V, E)$, where nodes V represent "locations" and (undirected) edges $E \subseteq V \times V$ represent connecting streets
- we call node sequence $v_0, v_1, \ldots, v_t$ with $(v_i, v_{i+1}) \in E \quad (i = 0, 1, \ldots, t-1)$ **walk in graph** G
- you know **starting node** but you do not know which walk your friend has taken in the graph
- idea: use a Markov chain to model **lack of knowledge** about current location (i.e. node) of your friend

Map of Würzburg

image credit: Map created from openstreetmap data

Random walks in graphs

- we can use Markov chain to model **random walk in graph** $G = (V, E)$ with nodes V and edges $E \subseteq V \times V$

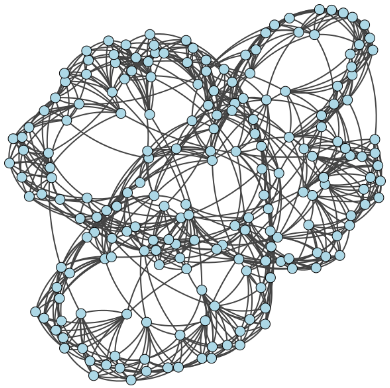
random walk as Markov chain

For a graph $G = (V, E)$, consider Markov chain model for random walks with discrete **state space** $E = V$.

- random variable $X_t \in V$ assumes state of the walker (i.e. visited node) at time $t \in \{0, 1, 2, \ldots\}$
- transitions must (generally) follow edges of graph, i.e. $P(X_t = v_t | X_{t-1} = v_{t-1}) > 0 \implies (v_{t-1}, v_t) \in E$
- any realization of the stochastic process X_t is a walk through graph G

- important class of models for **diffusion processes in networks** → Course: Statistical Network Analysis
- basis for **machine learning and recommender systems**

→ Course: Machine Learning for Complex Networks



random walk in an example graph

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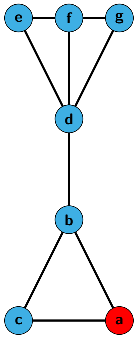
Adjacency matrices of graph



random walker with initial state $X_0 = a$

how can we define **transition matrix $\mathbf{T}$ of Markov chain** modelling random walk in G?

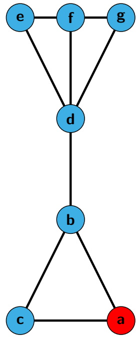
we can use entries A_{ij} of **adjacency matrix $\mathbf{A}$** to represent graph $G = (V, E)$, i.e.

$$A_{ij} = \begin{cases} 1 \text{ iff } (i,j) \in E \\ 0 \text{ else} \end{cases}$$

where i refers to row and j refers to column

$$\mathbf{A} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Transition matrix of random walk



random walker with initial state $X_0 = a$

how else could we define **transition probabilities**?

we can **define entries of transition matrix** as

$$T_{ij} := \frac{A_{ij}}{\sum_{k \in V} A_{ik}} = \frac{1}{d_i}$$

where T_{ij} is **probability of walker** in node i to move to node j and d_i is degree of node i

$$\mathbf{T} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

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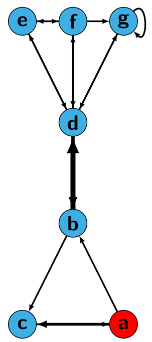
Biased random walks in weighted graphs



random walker with initial state $X_0 = a$

we can use edge weights to **bias transition probabilities**, e.g.

$$T_{ij} := \frac{A_{ij}}{\sum_{k \in V} A_{ik}}$$

$$\mathbf{T} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{8} & \frac{7}{8} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{4}{7} & 0 & 0 & \frac{1}{7} & \frac{1}{7} & \frac{1}{7} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix} \end{array}$$

Stochastic matrices

- for any current state i, corresponding **row in transition matrix is probability mass function** capturing $P(X_{t+1} = j | X_t = i)$ for all states j
- rows in the transition matrix $\mathbf{T}$ of a Markov chain must sum to one, i.e.

$$\sum_{j \in V} T_{ij} = 1 (\forall i \in V)$$

- we call such a matrix **row-stochastic** or **right-stochastic**
- let $\pi = (\pi_i)_{i \in V}$ be **stochastic row vector**, i.e. $\sum_i \pi_i = 1$
- $\pi \cdot \mathbf{T}$ exists and is again a stochastic row vector

$$\mathbf{T} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

example

$$\pi = (1, 0, 0, 0, 0, 0, 0)$$

$$\pi \cdot \mathbf{T} = \left(0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right)$$

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Visitation probabilities

- let $\pi^{(t)} = \left(\pi_1^{(t)}, \pi_2^{(t)}, \dots\right)$ be **probabilities of states** (visitation probabilities) at time t
- we call $\pi^{(0)}$ **initial state** of the Markov chain

example

$$\pi^{(0)} = (1, 0, 0, 0, 0, 0, 0)$$

$$\pi^{(1)} = \left(0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right) = \pi^{(0)} \cdot \mathbf{T}$$

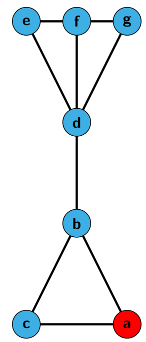
$$\pi^{(2)} = \left(\frac{5}{12}, \frac{1}{4}, \frac{1}{6}, \frac{1}{6}, 0, 0, 0\right)$$

$$= \pi^{(1)} \cdot \mathbf{T} = \pi^{(0)} \cdot \mathbf{T}^2$$

$$\mathbf{T} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

using initial state $\pi^{(0)}$ and transition matrix $\mathbf{T}$ we can calculate **state $\pi^{(t)}$ for any time t** as $\pi^{(t)} = \pi^{(0)} \cdot \mathbf{T}^t$

Total variation distance



- evolution of state visitation probabilities of Markov chain resembles a **diffusion process**
- diffundere (Latin) = **to spread out**, to **distribute**
- how can we quantify the difference between two states π and π'?

total variation distance

For two probability mass functions π and π' the **total variation distance** between π and π' is defined as

$$\delta\left(\pi, \pi'\right) := \frac{1}{2} \sum_i \left|\pi_i - \pi_i'\right| \in [0, 1]$$

- we can use $\delta\left(\pi^{(t)}, \pi^{(t+1)}\right)$ to quantify the **evolution of states** from time t to $t+1$

example

$$\pi^{(0)} = (1, 0, 0, 0, 0, 0, 0)$$

$$\pi^{(1)} = \left(0, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right)$$

$$\pi^{(2)} = \left(\frac{5}{12}, \frac{1}{4}, \frac{1}{6}, \frac{1}{6}, 0, 0, 0\right)$$

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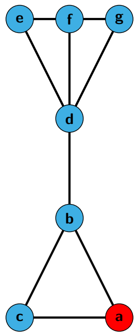
Group Exercise 14-01

- Consider a random walk in the graph on the right. For $\pi^{(0)} = (1, 0, \ldots)$ calculate the total variation distance between successive states $\pi^{(t)}$ and $\pi^{(t+1)}$ for different t.
- What do you observe in the evolution of $\delta(\pi^{(t)}, \pi^{(t+1)})$?



$$\mathbf{T} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

Stationary States of Markov chains

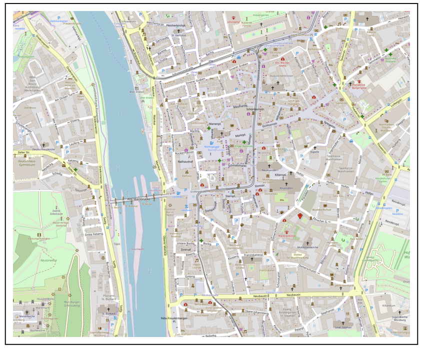
- in our example network, we observe

$$\delta\left(\pi^{(t)}, \pi^{(t+1)}\right) \to 0 \text{ for } t \to \infty$$

- in this example, Markov chain eventually reaches **stationary state**

$$\pi := \lim_{t\to} \pi^{(0)} \cdot \mathbf{T}^t$$

- after sufficiently long time, **it does not matter** at which time t we calculate visitation probabilities $\pi^{(t)}$
- allows us to make statements about **long-term behavior** of Markov chains (and random walks)



Map of Würzburg

image credit: Map created from openstreetmap data

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Stationary States and Eigenvalues

- Markov chain reaches **stationary state** $\pi := \lim_{t\to\infty} \pi^{(t)}$ such that

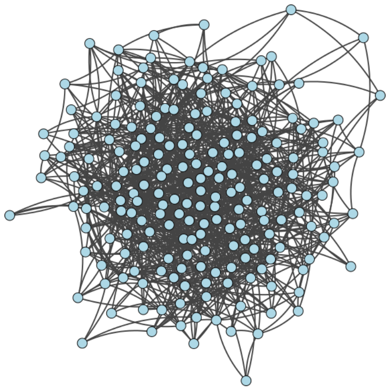
$$\pi = \pi \cdot \mathbf{T}$$

- this is an **eigenvalue problem** of the form

$$\pi \cdot \lambda = \pi \cdot \mathbf{T}$$

with eigenvalue $\lambda = 1$

- we can calculate stationary state as **left eigenvector of transition matrix corresponding to eigenvalue $\lambda = 1$**



convergence to stationary state for random walk in example graph

Practice Session

- we show how we can simulate random walks in graphs using `python`
- we calculate the total variation distance between stochastic vectors

practice session

see notebooks 14-01 and 14-02 in `gitlab` repository at

→ https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2

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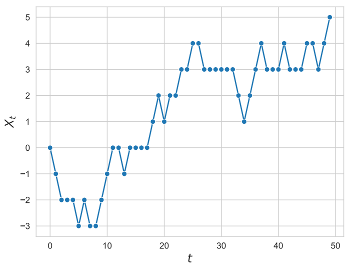
In summary

- we introduced **Markov chain models** for discrete-time categorical sequences
- key property of Markov processes is that they are **memoryless**, i.e. next state only depends on current state (and not on earlier steps in the trajectory)
- we can use Markov chains with **state space expansion** to model processes with any **finite memory** → self-study questions
- Markov chains are important class of probabilistic graphical models in **data science, machine learning, and theoretical physics**
- given transition matrix, how can we calculate **stationary state** of a Markov chain?



discrete-time stochastic process with **discrete state space**

Self-study questions

1. What is a stochastic process?
2. What is the Markov property and what is a Markov chain?
3. Consider a stochastic process, where we have a dependency of the next state to the current and the previous state, i.e. $P(X_{t+1} = s_{t+1} | X_t = s_t, X_{t-1} = s_{t-1})$. Can you change the state space such that you can use a Markov chain to model this process?
4. How can we define the transition matrix of a Markov chain capturing a random walk in an undirected graph?
5. Consider a random walk model, in which the random walker can 'restart' in an arbitrary node in each step with a fixed probability. For the example graph shown on slide 8, compute the transition matrix of such a random walk with restart probability $\alpha = 0.5$.
6. Does a random walk in a graph with symmetric adjacency matrix have a symmetric transition matrix?
7. What do you get if you multiply a row-stochastic matrix from the right to a stochastic row vector π, i.e. $\pi \cdot \mathbf{T}$?
8. How is total variation distance defined?
9. Give an example for two probability vectors where the total variation distance is maximal.
10. For a right-stochastic transition matrix $\mathbf{T}$ and an initial distribution $\pi^{(0)}$, how can we compute visitation probabilities of a random walk at time t?

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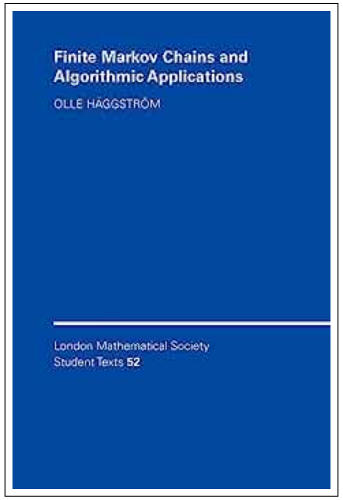
References

reading list

- AA Markov: **Extension of the limit theorems of probability theory to a sum of variables connected in a chain**, reprinted in Appendix B of *R. Howard: Dynamic Probabilistic Systems, volume 1: Markov Chains*, 1971
- O Häggström: **Finite Markov Chains and Algorithmic Applications**, 2002



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